3.5 Smooth Manifolds — Study Notes

These notes condense Lee, *Introduction to Smooth Manifolds* (2nd ed.), Chapters 1–9 and 11–17, with a glance at 13 and 19. "Smooth" means C^∞. Einstein summation is used where it helps: repeated upper and lower indices are summed.

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1. Smooth manifolds and smooth maps

1.1 Definitions

A **topological n-manifold** is a second countable Hausdorff space that is locally homeomorphic to $\mathbb{R}^n$. Hausdorff rules out the line with two origins; second countability guarantees partitions of unity and rules out the long line.

A **chart** is (U, φ) with U open and $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$ a homeomorphism onto an open set. Two charts are **smoothly compatible** if the **transition map** $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ is a diffeomorphism. A **smooth atlas** is a collection of pairwise compatible charts covering M. Each smooth atlas lies in a unique **maximal** atlas, the **smooth structure**.

A map $F : M \to N$ is **smooth** if for all charts, $\psi \circ F \circ \varphi^{-1}$ is smooth where defined. A **diffeomorphism** is a smooth bijection with smooth inverse.

Examples.

1. **Spheres.** $S^n \subset \mathbb{R}^{n+1}$ with stereographic charts $\sigma_\pm(x) = \dfrac{(x^1, \dots, x^n)}{1 \mp x^{n+1}}$. The transition map is $u \mapsto u/|u|^2$ on $\mathbb{R}^n \setminus 0$, smooth.
2. **Projective space.** $\mathbb{RP}^n = (\mathbb{R}^{n+1} \backslash 0)/\mathbb{R}^\times$, charts $U_i = \{x^i \neq 0\}$, $\varphi_i[x] = (x^0/x^i, \dots, \widehat{x^i/x^i}, \dots, x^n/x^i)$. Transitions are rational functions with non-vanishing denominators.
3. **Open subsets, products, graphs** of smooth functions; $GL(n, \mathbb{R})$ as an open subset of $\mathbb{R}^{n^2}$.
4. **Exotic structures.** On $\mathbb{R}$, the chart $x \mapsto x^3$ gives a smooth structure different from the standard one (the identity is not smooth in both directions), but the two manifolds are diffeomorphic via $x \mapsto x^{1/3}$. Genuinely non-diffeomorphic smooth structures exist on S^7 (Milnor) and on $\mathbb{R}^4$.

Manifolds with boundary use charts into the half-space $\mathbb{H}^n = \{x^n \geq 0\}$. Interior and boundary points are well defined (invariance of the boundary, by the inverse function theorem in the smooth case).

1.2 Partitions of unity

Bump functions. $f(t) = e^{-1/t}$ for $t > 0$ and 0 otherwise is smooth. From it build $h : \mathbb{R}^n \to [0, 1]$ smooth, equal to 1 on $\overline{B_1}$ and supported in B_2.

Theorem. For any open cover $\{U_\alpha\}$ of M there is a smooth partition of unity $\{\psi_\alpha\}$ subordinate to it: $\operatorname{supp} \psi_\alpha \subseteq U_\alpha$, the supports are locally finite, and $\sum_\alpha \psi_\alpha = 1$.

Consequences. Smooth Urysohn functions; extension of smooth functions from closed sets; existence of Riemannian metrics; existence of smooth exhaustion functions; Whitney embedding. **Partitions of unity are the main tool for passing from local to global** whenever the object being built is convex (metrics, inner products, connections) — but not for things like nonvanishing vector fields or forms, where averaging can produce zero.

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2. Tangent vectors and the differential

2.1 Derivations

A **tangent vector** at $p \in M$ is a **derivation** at p: a linear map $v : C^\infty(M) \to \mathbb{R}$ with $v(fg) = f(p)v(g) + g(p)v(f)$. The tangent space T_pM is the vector space of these. In a chart (x^i), the vectors

$$\left.\frac{\partial}{\partial x^i}\right|_p f = \frac{\partial (f \circ \varphi^{-1})}{\partial x^i}(\varphi(p))$$

form a basis, so $\dim T_pM = n$. (The proof uses Taylor's theorem with remainder $f(x) = f(p) + \sum (x^i - p^i) g_i(x)$ and the fact that derivations kill constants.)

Equivalent definitions: equivalence classes of curves through p with the same velocity in a chart; or vectors in $\mathbb{R}^n$ transforming by the Jacobian of transition maps. All are canonically isomorphic.

2.2 The differential

For smooth $F : M \to N$, $dF_p : T_pM \to T_{F(p)}N$ is $dF_p(v)(f) = v(f \circ F)$. In coordinates, dF_p has the Jacobian matrix $(\partial F^j / \partial x^i)$. **Chain rule:** $d(G \circ F)_p = dG_{F(p)} \circ dF_p$.

Change of coordinates. If $(\tilde{x}^j)$ is another chart,

$$\frac{\partial}{\partial x^i} = \frac{\partial \tilde{x}^j}{\partial x^i} \frac{\partial}{\partial \tilde{x}^j}.$$

The **tangent bundle** $TM = \bigsqcup_p T_pM$ is a smooth $2n$-manifold with charts (x^i, v^i) for $v = v^i \partial_i$; the projection $\pi : TM \to M$ is a smooth vector bundle of rank n. A **velocity** of a smooth curve γ is $\gamma'(t) = d\gamma(d/dt)$.

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3. Rank, immersions, submersions, submanifolds

3.1 The rank theorem

The **rank** of F at p is $\operatorname{rank} dF_p$. F is an **immersion** if dF_p is injective everywhere, a **submersion** if surjective everywhere. A **smooth embedding** is an immersion that is a homeomorphism onto its image (with the subspace topology).

Inverse function theorem. If dF_p is invertible, F is a diffeomorphism near p.

Constant rank theorem. If $F : M^m \to N^n$ has constant rank r near p, there are charts centered at p and $F(p)$ in which

$$F(x^1, \dots, x^m) = (x^1, \dots, x^r, 0, \dots, 0).$$

Proof sketch. Reorder so the top-left $r \times r$ block of the Jacobian is invertible; use the first r components of F plus the remaining x's as new coordinates on the source (IFT); then constant rank forces the remaining components of F to depend only on the first r, and a change of coordinates on the target straightens them.

Consequences. Immersions are locally embeddings. Submersions are open maps and have local sections.

Examples and counterexamples.

- The figure eight $\beta(t) = (\sin 2t, \sin t)$ on $(-\pi, \pi)$ is an injective immersion that is not an embedding: its image is compact but $(-\pi, \pi)$ is not, and points near $t = \pm\pi$ converge to $\beta(0)$ in the image.

- The **irrational line on the torus** $t \mapsto (e^{2\pi i t}, e^{2\pi i \alpha t}) \in T^2$, $\alpha \notin \mathbb{Q}$, is an injective immersion with dense image; not an embedding.
- $t \mapsto (t^2, t^3)$ is smooth and injective, a homeomorphism onto the cusp, but not an immersion at 0. Its image is not a smooth submanifold.
- An injective immersion of a **compact** manifold is an embedding (a continuous bijection from compact to Hausdorff is a homeomorphism).

3.2 Submanifolds

An **embedded submanifold** $S \subseteq M$ of dimension k is a subset that is a manifold in the subspace topology with a smooth structure making the inclusion an embedding. Equivalently, S satisfies the **local k-slice condition**: around each point of S there is a chart of M in which S is $\{x^{k+1} = \cdots = x^n = 0\}$.

Regular value theorem. Let $F : M^m \to N^n$ be smooth and $c \in N$ a **regular value**: dF_p is surjective for all $p \in F^{-1}(c)$. Then $F^{-1}(c)$ is an embedded submanifold of dimension $m - n$, and

$$T_p F^{-1}(c) = \ker dF_p.$$

More generally, the constant-rank level set theorem: if F has constant rank r, every level set is an embedded submanifold of codimension r.

Examples.

- $S^n = f^{-1}(1)$ for $f(x) = |x|^2$; $df_x(v) = 2x \cdot v$ is onto for $x \neq 0$. $T_x S^n = x^\perp$.
- $O(n) = \Phi^{-1}(I)$ for $\Phi : GL(n) \to \mathrm{Sym}(n)$, $\Phi(A) = A^T A$. $d\Phi_A(B) = B^T A + A^T B$; given $C \in \mathrm{Sym}(n)$, $B = \frac{1}{2} AC$ gives $d\Phi_A(B) = C$. So $O(n)$ is a submanifold of dimension $n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}$, with $T_I O(n) = \{B : B^T + B = 0\}$.
- $SL(n) = \det^{-1}(1)$; $d(\det)_A(B) = \det(A)\mathrm{tr}(A^{-1}B)$, so 1 is a regular value.

Sard's theorem. The set of critical values of a smooth map has measure zero. Hence regular values are dense.

Whitney embedding theorem. Every smooth n-manifold embeds in $\mathbb{R}^{2n}$ (easy version: $\mathbb{R}^{2n+1}$) and immerses in $\mathbb{R}^{2n-1}$ for $n > 1$.

Transversality. $F : M \to N$ is transverse to an embedded submanifold $S \subseteq N$ if $dF_p(T_p M) + T_{F(p)} S = T_{F(p)} N$ whenever $F(p) \in S$. Then $F^{-1}(S)$ is an embedded submanifold of codimension $\operatorname{codim} S$.

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4. Vector fields, flows, Lie brackets

4.1 Vector fields and flows

A **vector field** is a smooth section $X : M \to TM$. It acts on functions as a derivation $Xf(p) = X_p f$. An **integral curve** is γ with $\gamma'(t) = X_{\gamma(t)}$.

Fundamental theorem on flows. For smooth X there is a unique maximal flow $\theta : \mathcal{D} \to M$, with $\mathcal{D} \subseteq \mathbb{R} \times M$ open containing $\{0\} \times M$, such that $t \mapsto \theta_t(p)$ is the maximal integral curve through p, $\theta_0 = \mathrm{id}$, $\theta_s \circ \theta_t = \theta_{s+t}$ where defined, and each θ_t is a diffeomorphism between open sets. The proof is ODE existence, uniqueness and smooth dependence on initial conditions, plus a patching argument.

Completeness. Compactly supported vector fields, in particular all vector fields on compact manifolds, are complete (flow defined for all t). Counterexample: $X = x^2 \partial_x$ on $\mathbb{R}$ has flow $\theta_t(x) = x/(1 - tx)$, which blows up at $t = 1/x$.

Escape lemma. If an integral curve is defined only on a bounded-above interval, it leaves every compact set.

Canonical form. Near a point where $X_p \neq 0$ there are coordinates with $X = \partial/\partial x^1$ (straightening).

4.2 Lie bracket

$[X,Y]f = X(Yf) - Y(Xf)$. This is again a derivation (second derivatives cancel), so a vector field. In coordinates,

$$[X,Y] = \left(X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i} \right) \frac{\partial}{\partial x^j}.$$

Properties: bilinear, antisymmetric, Jacobi identity, $[fX, gY] = fg[X,Y] + f(Xg)Y - g(Yf)X$. Coordinate vector fields commute.

Naturality. If X is F-related to X' and Y to Y', then $[X,Y]$ is F-related to $[X',Y']$. In particular $F_*[X,Y] = [F_*X, F_*Y]$ for diffeomorphisms.

Lie derivative. $(\mathcal{L}_X Y)_p = \left.\frac{d}{dt}\right|_{t=0} d(\theta_{-t})_{\theta_t(p)}(Y_{\theta_t(p)})$. **Theorem:** $\mathcal{L}_X Y = [X,Y]$.

Commuting flows. $[X,Y] = 0$ iff $\theta^X_s \circ \theta^Y_t = \theta^Y_t \circ \theta^X_s$ wherever defined. More quantitatively, flowing around a small "square" of side $\sqrt{t}$ (along X, Y, $-X$, $-Y$) displaces a point by $t[X,Y] + O(t^{3/2})$.

Canonical form for commuting fields. If $X_1, \dots, X_k$ are linearly independent and pairwise commute near p, there are coordinates with $X_i = \partial/\partial x^i$.

4.3 Frobenius

A rank-k **distribution** $D \subseteq TM$ is **involutive** if $[X,Y] \in D$ whenever $X, Y \in D$, and **integrable** if each point lies in a k-dimensional submanifold tangent to D. **Frobenius theorem:** involutive $\Leftrightarrow$ integrable ($\Leftrightarrow$ locally of the form span $\{\partial_1, \dots, \partial_k\}$). Example of non-integrability: the contact distribution $\ker(dz - y\,dx)$ on $\mathbb{R}^3$.

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5. Tensors and differential forms

5.1 Covectors and tensors

$T^*_pM = (T_pM)^*$, with dual basis $dx^i|_p$. For $f \in C^\infty(M)$, $df_p(v) = vf$, so $df = \frac{\partial f}{\partial x^i} dx^i$. Covectors transform by $d\tilde{x}^j = \frac{\partial \tilde{x}^j}{\partial x^i} dx^i$.

Pullback. For $F : M \to N$ and a covector field ω on N, $(F^*\omega)_p(v) = \omega_{F(p)}(dF_p v)$. Pullback works for covariant tensors and forms even when F is not a diffeomorphism — unlike pushforward of vector fields.

Line integrals. $\int_\gamma \omega = \int_a^b \omega_{\gamma(t)}(\gamma'(t))\,dt$, independent of orientation-preserving reparametrization. **Fundamental theorem:** $\int_\gamma df = f(\gamma(b)) - f(\gamma(a))$. A 1-form is **exact** if $\omega = df$; **closed** if $d\omega = 0$, i.e. $\partial_i \omega_j = \partial_j \omega_i$. Exact $\Rightarrow$ closed; closed $\Rightarrow$ exact locally (on star-shaped sets), but not globally.

Key example. On $\mathbb{R}^2 \setminus 0$,

$$\omega = \frac{x\,dy - y\,dx}{x^2 + y^2}$$

is closed, but $\int_{S^1} \omega = 2\pi \neq 0$, so it is not exact. Locally $\omega = d\theta$ for a branch of the angle function.

A covariant k-tensor field assigns a multilinear map $(T_pM)^k \to \mathbb{R}$ to each p. A **Riemannian metric** is a symmetric positive definite 2-tensor field $g = g_{ij}dx^i dx^j$.

5.2 Alternating tensors and the wedge product

$\Lambda^k(V^*)$ is the space of alternating k-linear forms on V; $\dim \Lambda^k(V^*) = \binom{n}{k}$ with basis $\varepsilon^{i_1} \wedge \cdots \wedge \varepsilon^{i_k}$, $i_1 < \cdots < i_k$. The wedge product (Lee's determinant convention) satisfies

$$\varepsilon^{i_1} \wedge \cdots \wedge \varepsilon^{i_k}(v_1, \dots, v_k) = \det\big(\varepsilon^{i_a}(v_b)\big),$$

is associative, bilinear, and **graded commutative**: $\omega \wedge \eta = (-1)^{k\ell}\eta \wedge \omega$. In particular $\omega \wedge \omega = 0$ for odd k. For n-forms on an n-dimensional V and a linear map T, $T^*\omega = (\det T)\,\omega$.

Pitfall. Conventions differ by factorials (Lee/Spivak "determinant" convention vs Kobayashi–Nomizu "Alt" convention). Fix one and stick to it.

Interior product. $(i_X\omega)(v_2, \dots, v_k) = \omega(X, v_2, \dots, v_k)$. It is an antiderivation: $i_X(\omega \wedge \eta) = (i_X\omega) \wedge \eta + (-1)^k \omega \wedge i_X\eta$.

5.3 The exterior derivative

Theorem. There is a unique family of linear maps $d : \Omega^k(M) \to \Omega^{k+1}(M)$ such that: (i) df is the differential for functions; (ii) $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$; (iii) $d \circ d = 0$. In coordinates,

$$d\Big(\sum_I \omega_I dx^I\Big) = \sum_I \sum_j \frac{\partial \omega_I}{\partial x^j} dx^j \wedge dx^I.$$

$d \circ d = 0$ is equality of mixed partials. **Naturality:** $F^*(d\omega) = d(F^*\omega)$.

Invariant formula for 1-forms: $d\omega(X, Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X, Y])$.

In $\mathbb{R}^3$ d on 0-, 1-, 2-forms is grad, curl, div, and $d^2 = 0$ encodes $\operatorname{curl}\operatorname{grad} = 0$ and $\operatorname{div}\operatorname{curl} = 0$.

Lie derivative of forms and Cartan's formula.

$$\mathcal{L}_X\omega = \frac{d}{dt}\Big|_{t=0} \theta_t^*\omega = i_X(d\omega) + d(i_X\omega).$$

Proof idea. Both sides are derivations of degree 0 commuting with d and agreeing on functions ($\mathcal{L}_X f = Xf = i_X df$); such operators are determined by their values on functions and on df, and the forms $f\,dg$ generate locally. Corollary: $\mathcal{L}_X$ commutes with d.

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6. Orientation and integration

6.1 Orientation

An **orientation** of an n-dimensional vector space is an equivalence class of ordered bases under "change of basis matrix has positive determinant". A pointwise orientation on M is **continuous** if locally given by a local frame. M is **orientable** iff any of the following equivalent conditions holds:

- there is a nowhere-vanishing n-form (a **volume form**);
- there is an atlas whose transition maps have positive Jacobian determinant;
- (for connected M) the orientation double cover is disconnected.

Examples: spheres, tori, Lie groups (left-translate a nonzero element of $\Lambda^n(T^*_eG)$), and any hypersurface of an orientable manifold (e.g. of $\mathbb{R}^n$) with a nowhere-vanishing normal field are orientable. $\mathbb{RP}^n$ is orientable iff n is odd. The Möbius band is not orientable. Every simply connected manifold is orientable.

Boundary orientation. If M is oriented with boundary, ∂M gets the **Stokes orientation**: $(v_1, \dots, v_{n-1})$ is positive on ∂M iff $(N, v_1, \dots, v_{n-1})$ is positive on M, where N is outward-pointing. Equivalently, if ω orients M, then $i_N\omega$ orients ∂M. For $\mathbb{H}^n = \{x^n \geq 0\}$ with the standard orientation, the induced orientation on $\partial\mathbb{H}^n \cong \mathbb{R}^{n-1}$ is $(-1)^n$ times the standard one.

6.2 Integration

For an n-form $\omega = f\,dx^1 \wedge \cdots \wedge dx^n$ compactly supported in a positively oriented chart, $\int_M \omega = \int_{\mathbb{R}^n} f\,dx^1 \cdots dx^n$. This is well defined by the change of variables formula: under an orientation-preserving diffeomorphism the Jacobian $|\det DF| = \det DF$ matches the transformation $F^*(dy^1 \wedge \cdots \wedge dy^n) = \det(DF)dx^1 \wedge \cdots \wedge dx^n$. In general, patch with a partition of unity.

Properties: linear; reverses sign under reversing orientation; $\int_M F^*\omega = \pm\int_N \omega$ for a diffeomorphism F preserving/reversing orientation; positive on positively oriented volume forms.

6.3 Stokes' theorem

Theorem (Stokes). Let M be an oriented smooth n-manifold with boundary, ω a compactly supported $(n-1)$-form. Then

$$\int_M d\omega = \int_{\partial M} \omega,$$

with ∂M carrying the Stokes orientation (if $\partial M = \emptyset$, the right side is 0).

Proof sketch. By a partition of unity, reduce to ω supported in one chart. For $\mathbb{H}^n$ write $\omega = \sum_i \omega_i\,dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$, so $d\omega = \sum_i (-1)^{i-1}\partial_i\omega_i\,dx^1 \wedge \cdots \wedge dx^n$. Integrate each term with Fubini: for $i < n$ the integral of $\partial_i\omega_i$ in x^i is 0 (compact support); for $i = n$ it gives $-\omega_n(x', 0)$. So $\int_{\mathbb{H}^n} d\omega = (-1)^n \int_{\mathbb{R}^{n-1}} \omega_n(x', 0)\,dx'$, which is exactly $\int_{\partial\mathbb{H}^n}\omega$ with the orientation $(-1)^n\times$standard. □

Special cases. Fundamental theorem of calculus ($n = 1$); Green's theorem; divergence theorem $\int_M \operatorname{div} X\,dV = \int_{\partial M}\langle X, N\rangle\,dA$; classical Stokes for surfaces in $\mathbb{R}^3$.

Corollaries.

- On a compact oriented manifold without boundary, $\int_M d\eta = 0$. So a volume form is never exact, and $H^n_{dR}(M) \neq 0$.
- If ω is a closed k-form and S_0, S_1 are homologous compact oriented k-submanifolds (cobounding a compact oriented $(k+1)$-manifold), then $\int_{S_0}\omega = \int_{S_1}\omega$.
- **Smooth Brouwer:** there is no smooth retraction $r : \overline{B}^n \to S^{n-1}$. If there were, let σ be a volume form on S^{n-1} with $\int\sigma > 0$; then $0 < \int_{S^{n-1}}\sigma = \int_{S^{n-1}} r^*\sigma = \int_{B^n} d(r^*\sigma) = \int_{B^n} r^*(d\sigma) = 0$, since $d\sigma$ is an n-form on an $(n-1)$-manifold.

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7. de Rham cohomology

$$H^k_{dR}(M) = \frac{\ker\left(d : \Omega^k(M) \to \Omega^{k+1}(M)\right)}{\operatorname{im}\left(d : \Omega^{k-1}(M) \to \Omega^k(M)\right)}.$$

It is a contravariant functor, and the wedge product makes $H^*_{dR}(M)$ a graded-commutative ring.

Basic facts.

- $H^0_{dR}(M) = \mathbb{R}^{\#\text{components}}$ (closed functions are locally constant).
- $H^k_{dR}(M) = 0$ for $k > \dim M$.
- **Homotopy invariance.** Homotopic smooth maps induce the same map; proof by a chain homotopy h with $dh + hd = g^* - f^*$, built from integrating along the homotopy parameter. Continuous maps can be approximated by smooth ones, so H^*_{dR} is a homotopy invariant of manifolds.
- **Poincaré lemma.** Star-shaped open sets in $\mathbb{R}^n$ have $H^k_{dR} = 0$ for $k \geq 1$. Every closed form is locally exact.
- **Mayer–Vietoris.** For $M = U \cup V$ open,

  $$0 \to \Omega^k(M) \xrightarrow{(i_U^*,\, i_V^*)} \Omega^k(U) \oplus \Omega^k(V) \xrightarrow{\omega - \eta} \Omega^k(U \cap V) \to 0$$

  is exact (surjectivity uses a partition of unity $\{\varphi_U, \varphi_V\}$: $\omega = (\varphi_V \omega) - (-\varphi_U \omega)$), giving a long exact sequence with connecting map $\delta : H^k(U \cap V) \to H^{k+1}(M)$.

Computations.

- $H^k_{dR}(S^n) = \mathbb{R}$ for $k = 0, n$ and 0 otherwise ($n \geq 1$).
- $H^*_{dR}(T^n) = \Lambda^*(\mathbb{R}^n)$, generated by $dx^1, \dots, dx^n$.
- $H^1_{dR}(\mathbb{R}^2 \setminus 0) = \mathbb{R}$, generated by the angle form.
- For M compact, connected, oriented, without boundary, integration gives $H^n_{dR}(M) \cong \mathbb{R}$. If M is compact connected and non-orientable, $H^n_{dR}(M) = 0$.

Degree. For a smooth map $F : M \to N$ between compact connected oriented n-manifolds without boundary, $\int_M F^*\omega = \deg(F) \int_N \omega$, where $\deg F$ is the signed count of preimages of any regular value. This agrees with the homological degree of 3.4.

de Rham theorem. $H^k_{dR}(M) \cong H^k(M; \mathbb{R})$ (singular cohomology with real coefficients), and the isomorphism is a ring isomorphism (wedge $\leftrightarrow$ cup). The map is integration of forms over smooth singular chains, via Stokes.

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8. Riemannian metrics (introduction)

A **Riemannian metric** g is a smoothly varying inner product on each T_pM. **Existence:** on each chart take the Euclidean metric, then glue with a partition of unity; positive-definiteness survives convex combinations.

Lengths and distance. $L(\gamma) = \int_a^b |\gamma'(t)|_g \, dt$; $d_g(p,q) = \inf L(\gamma)$ over piecewise smooth curves. d_g is a metric inducing the manifold topology.

Musical isomorphisms. g gives $TM \cong T^*M$: $X^\flat = g(X, \cdot)$, i.e. $(X^\flat)_j = g_{ij} X^i$; inverse $\sharp$ uses g^{ij}. **Gradient:** $\operatorname{grad} f = (df)^\sharp$, so $g(\operatorname{grad} f, X) = Xf$; in coordinates $\operatorname{grad} f = g^{ij} \partial_j f \, \partial_i$.

Riemannian volume form. On an oriented Riemannian manifold there is a unique n-form dV_g with $dV_g(E_1, \dots, E_n) = 1$ on positive orthonormal frames; in coordinates $dV_g = \sqrt{\det(g_{ij})}\, dx^1 \wedge$

$\cdots \wedge dx^n$. **Divergence** is defined by $d(i_X dV_g) = (\operatorname{div} X)\, dV_g$, and Stokes becomes the divergence theorem.

Isometries preserve g. Examples: round metric on S^n (induced from $\mathbb{R}^{n+1}$), flat metric on T^n, hyperbolic metric $\frac{dx^2+dy^2}{y^2}$ on the upper half-plane. Curvature, connections and geodesics are the subject of the Riemannian geometry elective.

8A. Lie groups and vector bundles (light)

Lie groups. A Lie group G is a smooth manifold with smooth multiplication and inversion. Examples: $\mathbb{R}^n$, T^n, $GL(n,\mathbb{R})$, and its closed subgroups $O(n)$, $SO(n)$, $SL(n,\mathbb{R})$, $U(n)$, $SU(n)$ (all embedded submanifolds by the regular value theorem; in general by the **closed subgroup theorem**: every closed subgroup of a Lie group is an embedded Lie subgroup).

A vector field X on G is **left-invariant** if $d(L_g)(X_h) = X_{gh}$ for all g, h. Evaluation at e is a linear isomorphism from left-invariant fields to T_eG, and the bracket of left-invariant fields is left-invariant (naturality of the bracket). This makes $\mathfrak{g} = T_eG$ a **Lie algebra**. For $G = GL(n,\mathbb{R})$, $\mathfrak{g} = M_n(\mathbb{R})$ with $[A,B] = AB - BA$. Consequences:

- **Every Lie group is parallelizable** (left-translate a basis of T_eG), hence orientable. So S^2 admits no Lie group structure.
- Left-invariant fields are complete; the integral curve through e of the field with $X_e = A$ is a one-parameter subgroup, and $\exp(A)$ is its value at $t = 1$. For matrix groups, $\exp(A) = e^A = \sum A^k/k!$.

Vector bundles. A smooth rank-k vector bundle $\pi : E \to M$ has fibres $E_p \cong \mathbb{R}^k$ and local trivializations $\pi^{-1}(U) \cong U \times \mathbb{R}^k$ that are linear on fibres. TM, T^*M, $\Lambda^k T^*M$ and tensor bundles are all vector bundles; their smooth sections are vector fields, 1-forms, k-forms and tensor fields. A bundle is **trivial** iff it has a global frame (k sections that are linearly independent at every point). The tautological line bundle over $\mathbb{RP}^1 \cong S^1$ is the open Möbius band, the simplest non-trivial bundle. Transition functions $g_{\alpha\beta} : U_\alpha \cap U_\beta \to GL(k,\mathbb{R})$ satisfy the cocycle condition $g_{\alpha\beta} g_{\beta\gamma} = g_{\alpha\gamma}$ and determine the bundle; for TM they are the Jacobians of the chart transitions, and M is orientable iff they can be chosen with positive determinant.

8B. Worked example: the torus

Let $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ with coordinates (x, y) defined mod 1. The forms dx, dy descend to T^2 (translation-invariant) and are closed; they are not exact, since $\int_a dx = 1$ along the loop $a(t) = (t, 0)$, and exact forms integrate to 0 over loops. The 2-form $dx \wedge dy$ is a volume form with $\int_{T^2} dx \wedge dy = 1$.

Claim: $H^1_{dR}(T^2)$ has basis $[dx], [dy]$. Given a closed 1-form ω, let $a = \int_a \omega$, $b = \int_b \omega$ over the two standard loops. Then $\eta = \omega - a\, dx - b\, dy$ is closed with zero integrals over a and b. Pull η back to $\mathbb{R}^2$: there it is exact, $\eta = df$ (Poincaré lemma). The periods vanish, so $f(x+1, y) = f(x, y)$ and $f(x, y+1) = f(x, y)$ (the difference is the integral of η over a translate of a or b, which is homotopic to a or b). So f descends to T^2 and $[\omega] = a[dx] + b[dy]$. Independence follows by integrating over a and b. Together with $H^0 = \mathbb{R}$ and $H^2 = \mathbb{R}$ (compact, connected, orientable), $H^*_{dR}(T^2) = \Lambda^*(\mathbb{R}^2)$, with $[dx] \wedge [dy]$ generating H^2. This matches the cohomology ring of T^2 in 3.4 §5.2, as de Rham's theorem predicts.

Degree computation. The map $F : T^2 \to T^2$ induced by an integer matrix $A = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$

satisfies $F^*(dx \wedge dy) = (ps - qr)\, dx \wedge dy$, so $\deg F = \det A$. Check with regular values: F is a $|\det A|$-sheeted covering when $\det A \neq 0$, each sheet contributing sign $\det A$.

9. Common pitfalls

1. **Bijective immersion ≠ embedding; homeomorphism onto image ≠ embedding.** Check both the immersion condition and the topology.
2. **Images vs preimages.** The regular value theorem is about preimages. Images of smooth maps need not be submanifolds (cusp, figure eight).
3. **Pushforward of vector fields** is only defined for diffeomorphisms (or for F-related fields). Pullback of forms always works.
4. **Closed ≠ exact.** The topology of M is precisely the obstruction, measured by H^*_{dR}.
5. **Orientation signs in Stokes.** "Outward normal first." Check on $[0,1]$: $\partial[0,1] = \{1\} - \{0\}$.
6. **Wedge product conventions** vary by factorials between books; check $dx \wedge dy(e_1, e_2) = 1$ in your convention.
7. d **of a coefficient-free basis form is zero only in coordinates.** $d(dx^i) = 0$, but for a general frame ε^i, $d\varepsilon^i$ need not vanish (it encodes the bracket of the dual frame).
8. **Completeness.** Don't assume flows exist for all time on non-compact manifolds.
9. **Partitions of unity can't average away topology.** You can glue metrics but not nonvanishing vector fields (S^2 has none).

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10. Connections forward

- **Algebraic topology (3.4).** de Rham theorem; Poincaré duality $H^k_{dR}(M) \cong H^{n-k}_{dR}(M)^*$ for compact oriented M (via the pairing $\int \alpha \wedge \beta$); degree theory.
- **Riemannian geometry (elective).** Connections, geodesics, curvature, Gauss–Bonnet $\int_M K\, dA = 2\pi\chi(M)$, linking the metric to the topology computed in 3.4.
- **Lie groups (elective).** Left-invariant vector fields, exponential map (a flow), Lie algebra = bracket of left-invariant fields, closed subgroup theorem.
- **Functional analysis and PDE (3.2, PDE elective).** Hodge theory: on compact oriented Riemannian M each de Rham class has a unique harmonic representative, proved with elliptic PDE.
- **Complex analysis (2.2).** Cauchy's theorem is Stokes for the closed form $f(z)\, dz$; residues compute H^1_{dR} classes of punctured domains.
- **Physics.** Maxwell's equations as $dF = 0$, $d{\star}F = J$; Hamiltonian mechanics on symplectic manifolds $(T^*M, d\lambda)$.